

Misoprostol in the treatment of tinnitus: A double-blind study

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OBJECTIVE: To test the efficacy of misoprostol as a treatment for tinnitus.

DESIGN: A prospective, placebo-controlled, double-blind study.

SETTING: Başkent University Otolaryngology Clinic.

PATIENTS: Forty adult patients who had had tinnitus for a minimum of 6 months and were free of systemic or otolaryngologic disease. Twenty-eight patients were randomly assigned to the experimental group (group I) and 12 to the control group (group II).

INTERVENTION: The respective groups received active drug and placebo in increasing doses for 4 months. The effect of medications on tinnitus were evaluated by determining improvement rates in tinnitus loudness and subjective tinnitus scoring.

RESULTS: In the experimental group, 18 of 28 patients showed improvement in tinnitus loudness, representing an improvement rate of 64%. The improvement rate based on subjective tinnitus scoring was 36% (10 of 28 patients). In the control group, the improvement rate for tinnitus loudness was 33% ($n = 4$), and the rate for subjective tinnitus scoring was 17% ($n = 2$). The difference between improvement rate for tinnitus loudness of the experimental group and control group was found to be statistically significant ($P = 0.039$), but difference between improvement rate based on subjective tinnitus scoring was insignificant ($P = 0.119$). When results in the experimental group were analyzed according to etiological factors, the improvement rate was highest in the sudden-onset subgroup (77%).

CONCLUSIONS: Misoprostol provided therapeutic relief for some patients with tinnitus we studied, but further investigation of larger groups is needed. (Otolaryngol Head Neck Surg 2004;130:604-10.)

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Tinnitus is defined as the conscious experience of a sound that originates in the head.¹ Affecting roughly 17% of the overall population and 33% of the elderly population, it is one of the most common symptoms of the hearing system.¹ Because of its psychological effects, tinnitus can severely reduce quality of life for patients and their families, and some reported cases have even involved suicide attempts. Some types of tinnitus that affect a significant proportion of the population and cause psychological stress can be objectively detected and treated. However, many patients have a purely subjective form. To date, there is no effective treatment for these patients, and subjective tinnitus is a challenge faced by many ear-nose-throat (ENT) physicians.

This study focused on subjective tinnitus. Medical treatment for this condition was introduced in the mid-20th century. Though many therapeutic modalities have been attempted, no adequate solution has been found. The first attempt to treat subjective tinnitus with misoprostol was made in 1993.² This trial was based on studies after the 1960s that investigated prostaglandins (PGs), salicylate ototoxicity, and cochlear metabolism.³ Various reports documented that 3 different prostaglandins (PGI_2 , PGE_2 , and PGF_2) were synthesized in the stria vascularis and the spiral ligament, and also that PGI_2 and PGF_2 were present in perilymph.⁴⁻⁶ It was shown that PGs, acting as neuromodulators, affected the afferent pathway in the cochlea, and played a role in the hormonal control of cochlear microcirculation through vasodilator effects of PGI_2 and PGE_2 , and the vasoconstrictor effect of PGF_2 .^{7,8}

We designed a prospective, placebo-controlled, double-blind study to test the efficacy of misoprostol (Cytotec), a synthetic prostaglandin E_1 analogue, in the treatment of tinnitus. Objective and subjective criteria were used to assess improvement, and results in experimental and placebo (control) groups were compared.

MATERIAL AND METHOD

The study protocol was approved by Ethical Committee of Başkent University Hospital, Ankara and Turkish Republic Health Ministry. Before study, all subjects were informed about possible side effects and were asked to complete and sign the clinic research volunteer subject information and consent form. We included 40 patients who presented to the ENT Department of the Faculty of Medicine of Başkent University with the complaint of tinnitus between June 2000 and June 2001. The inclusion criteria were as follows: 6 months minimum tinnitus duration; normal ENT examination; no systemic disease (hypertension, diabetes, hyperlipidemia, hypo-hyperthyroidism); no ENT pathology that could induce tinnitus (Meniere's disease, objective causes of tinnitus, otosclerosis, among others); no known neuropsychiatric disease; and no pregnancy.

Patients who met these criteria underwent complete blood count and biochemical testing in addition to a routine ENT examination. Further investigations with electronystagmography (ENG) and internal acoustic magnetic resonance imaging were performed when indicated. The subjects were informed about tinnitus and the appropriate treatment, and they voluntarily signed an informed consent form.

Audiological examinations were performed before the start of treatment and every month thereafter. The audiometrists performing these examinations were blinded to the treatment received by the patient. Each of these examinations included audiological checks, high-frequency audiometry, and impedancemetric tests. For the audiological checks and high-frequency audiometry, we used the Clinical Audiometer AC40 with Telephonics TDH-39P headphones for low frequency and KOSS HV/PRO Digital headphones for high frequency. To record impedancemetric measurements, we used the Impedance Audiometer AZ7. (Interacoustics Co. Assens, Denmark). Airway hearing thresholds were determined by routine audiological examination at 0.125 to 8 kHz, and by high-frequency audiometry examination at 10 to 18 kHz. The hearing test results were evaluated for degree of hearing loss on the basis of pure tone average (500+1000+2000 Hz) in low frequencies.⁹

Each patient's tinnitus frequency and loudness were determined at the end of every month. The pitch-match frequency and loudness match -dB SL were used for determination of frequency and loudness. Patients were also asked to grade the loudness of their tinnitus. In this evaluation, one of the participant physicians asked the patient to what extent the heard noise annoyed him or her and requested the patient to assess the degree of irritation using a 10-point scoring system (0 = I do not hear and am not annoyed; 10 = The voice I hear makes me extremely annoyed). At each examination, every subject also filled in a small questionnaire that inquired whether the tinnitus caused any problems with sleeping or inattentiveness during daily activities.

The 40 patients were randomly assigned to 1 of 2 groups. Cytotec tablets were repackaged in labeled bottles such that the packaging and tablets looked identical to the placebos. In order to be able to avoid any side effects, patients in group I (n = 28) were initially given 200 µg/day (1 tb/day) of misoprostol in the first week. This was then increased by 200 µg/day each subsequent week. The dose of 800 µg/day (4 × 1 tb/day) used commonly in gastroenterologic diseases was reached at the end of the first month, and this dosage was continued for 3 more months. The patients in group II (n = 12) received placebo tablets (each containing 75% lactose, 25% cellulose, and Mg-stearate up to 1% of the total weight) on the same dosing schedule.

None of the participants received any additional medication during the course of the study. They were requested to consult the investigators if any possible side effects occurred.

For each patient, the dosage of the medications and the results of audiological investigations were recorded on the "Form of Misoprostol Use in Tinnitus," which was started when the patient was first enrolled in the study. This form also included sections on tinnitus frequency and loudness, subjective tinnitus scores, sleep pattern, attentiveness, side effects, as well as patient demographic characteristics, detailed description about the individual's tinnitus problems, etiological factors, and medical history.

Improvement was assessed on a case-by-case basis, and the mean values were used to analyze the results. Two major criteria of improvement were used. The first was determination of a 15-dB or greater decrease from baseline tinnitus loudness after 4 months of treatment (that is, at least 15 dB reduction in tinnitus loudness over 4 months). The second criterion was determination of reduction of 3 or more grades on tinnitus scoring from baseline after 4 months (that is, a drop of at least 3 grades on the 10-point scale rating level of annoyance from tinnitus). In addition to these major criteria, we assessed the sleep patterns and attentiveness as minor subjective criteria. Tinnitus frequency was not used as an improvement criteria. The reasons for this were that our analysis revealed day-to-day and within-day variation while other parameters remained unchanged. This finding is in accordance with other authors' findings that reliability of pitch-match frequency varies from patient to patient.¹

For statistical testing, we used analysis of variance for repeated measurements (repeated measurements of the overall linear model) to compare the 5 recordings (baseline and months 1 through 4) of tinnitus frequency and loudness, subjective scores, sleep pattern, and attentiveness in the placebo and treatment groups. Chi-square testing was used to analyze group differences in the distribution of categorical data. The Mann-Whitney-U test was used to compare nonparametric data between the 2 groups, and parametric data were compared using the Student's *t* test. The paired-samples *t* test was used to assess for significant differences among paired samples within each group. We used the Levene test to evaluate the homogeneity of the variables in each group, (*P* values <0.05 were considered to indicate statistical significance).

RESULTS

Of the 40 patients, 14 were females and 26 were males. The mean patient age was 51.17 ± 13.14 years (range, 19 to 70 years). Group I included 8 females and 20 males of mean age 51.43 ± 12.95 years (range, 19 to 70 years). Thirteen of the patients were affected bilaterally, and 15 had unilateral tinnitus. The right ear was involved in 21 cases, and the left ear in 20 cases. The mean duration of tinnitus was 39.39 ± 34.30 months

(range, 6 to 120 months). Tinnitus was continuous in 24 patients and intermittent in 4 cases. Thirteen of the patients reported sudden onset, 6 had experienced acoustic trauma, 4 had developed ototoxicity (3 due to aspirin and 1 due to aminoglycosides), 2 had sustained mechanical trauma, and 3 had a history of ENT surgery (septoplasty). Fifteen of the patients developed side effects, including nausea (*n* = 8), abdominal pain (*n* = 6), and diarrhea (*n* = 1).

Group II included 6 females and 6 males of mean age 50.58 ± 14.16 years (range, 23 to 70 years). Four of the patients were affected bilaterally, and 8 had unilateral tinnitus. The right ear was involved in 8 cases, and the left ear in 8 cases as well. The mean duration of tinnitus was 26.08 ± 21.32 months (range, 6 to 72 months). Tinnitus was continuous in 8 patients and intermittent in 4 cases. Two patients reported sudden onset, 2 had suffered acoustic trauma, 4 had developed ototoxicity (3 due to aspirin and 1 due to aminoglycosides), and 3 had sustained mechanical trauma. The side effects in the placebo-treated group were nausea (*n* = 1) and diarrhea (*n* = 3).

When the above group findings were compared, the only statistical difference was in the distribution of observed side effects. There was a notable difference in the distribution of sudden-onset cases between the 2 groups, but the level was not statistically significant (*P* = 0.07).

In addition to the above parameters, the groups were similar regarding age, gender, distribution of profession, distribution of tinnitus type (continuous or intermittent); tinnitus duration; numbers of patients with acoustic trauma or history of vascular disease; tinnitus frequency, loudness, and subjective tinnitus score, sleep pattern and attentiveness of the patients; occurrence of side effects; and numbers of patients with bilateral tinnitus. We evaluated the 40 patients' pure tone averages according to the system of Stach (1998),⁹ and these data are listed in Table 1.

Comparison of tinnitus frequencies. Analysis showed that frequency change was related in some degree to time, but the association was not statistically significant (*P* = 0.075). The interaction between the 2 groups was significant (*P* = 0.029). There were no significant differences in

Table 1. Low-frequency pure tone averages in the 40 patients studied^a

	Right ear (%)	Left ear (%)
Normal hearing (−10 to 15 dB)	18 (45)	21 (52.5)
Very mild HL (16-25 dB)	14 (35)	8 (20)
Mild HL (26-40 dB)	6 (15)	8 (20)
Moderate HL (41-55 dB)	1 (2.5)	3 (7.5)
Moderately severe HL (56-70 dB)	0 –	0 –
Severe HL (71-90 dB)	1 (2.5)	0
Profound HL (91+ dB)	0 –	0 –

HL, Hearing loss

the frequency data recorded for the 2 groups at any of the testing stages ($P = 0.18$).

Comparison of tinnitus loudness. The influence of time on the loudness change was observed to be statistically significant ($P=0.001$). The interaction between the 2 groups was observed to be insignificant ($P = 0.52$). No significant difference was observed on the loudness data between the 2 groups ($P = 0.13$).

Comparison of tinnitus scores. The influence of time on the score change was observed to be statistically significant ($P = 0.001$). The interaction between the 2 groups was observed to be insignificant ($P = 0.66$). No significant difference was observed on the score data between the 2 groups ($P = 0.34$).

Comparison of sleep patterns. In group I, 20 of the patients had a regular sleep pattern and 8 reported delayed onset of sleep. In group II, 9 patients had a regular pattern and 3 reported delayed onset. There was no significant difference between the groups regarding sleep pattern at pre-treatment evaluation ($\chi^2 = 0.05$, $P = 1$) and the patterns remained unchanged throughout the study.

Comparison of attentiveness. In group I, 23 of the patients reported a normal level of attentiveness throughout the day, 4 were slightly affected but maintained regular activity, and 1 exhibited loss of attentiveness to a degree where normal tasks could not be completed. In group II, 9 patients reported a normal level of attentiveness throughout the day, and 3 reported being slightly affected but able to maintain regular activity. There was no significant difference between the groups regarding level of attentiveness at pretreatment evaluation ($\chi^2 = 1.03$, $P = 0.59$), and no

changes in attentiveness were noted at any of the other test times.

Improvement Rates

Evaluation of tinnitus loudness. In group I, 18 of the 28 patients exhibited improvement at the end of 4 months, representing an improvement rate of 64%. In this group, 46% ($n = 13$) of patients showed 20 dB or greater improvement in the objective tinnitus assessment. In group II, 4 (33%) of the 12 patients exhibited improvement in tinnitus loudness. When t testing relating to the difference between rates was performed, the difference between improvement rate of group I and group II was found to be statistically significant ($P = 0.039$).

Evaluation of subjective tinnitus scores. In group I, 10 of the 28 patients showed improvement based on subjective scoring, representing an improvement rate of 36%. In group II, 2 (17%) of the 12 patients exhibited subjective improvement in tinnitus. When t testing relating to difference between rates was performed, the difference between improvement rate of group I and group II was not found to be statistically significant ($P = 0.119$).

Improvement rates in relation to etiological factors. When the group I patients were grouped according to the etiological factors in tinnitus, those with a history of sudden onset showed the highest rate of improvement (Table 2). At the end of 4 months of treatment, the tinnitus symptoms were totally eliminated in 2 patients in group I and 1 patient in group II. Although not statistically significant ($P > 0.05$), this result was considered to be an important sign of the efficacy of misoprostol.

Table 2. Distribution of improvement rates relative to etiological factors in the misoprostol-treated group

Etiological factor	Improvement in tinnitus loudness (%)	Improvement in tinnitus scores (%)
Sudden onset (n = 13)	10 (77)	4 (30)
Acoustic trauma (n = 6)	4 (66)	2 (33)
Mechanical trauma/surgery (n = 5)	1 (20)	2 (40)
Ototoxicity (n = 4)	2 (50)	2 (50)
Total (N = 28)	18	10

DISCUSSION

Tinnitus is most often observed between the ages of 40 and 80 years.¹⁰ The mean age of the subjects we studied was 51.2 years, which is in accord with this reported range. Both genders are affected by tinnitus at equal rates.¹⁰ The gender distribution in our study was 35% females and 65% males (male/female = 1.86). These findings are not significantly different from the 50% male to 50% female proportions noted in the literature ($\chi^2 = 0.92$, $P = 0.33$). The imbalance of the genders in our investigation was due to the limited number of patients who were available (met the criteria) for the study. Regarding other demographic features, tinnitus is bilateral in 50% of patients,¹⁰ and the 42.5% bilateral cases we observed is in accord with this.

The distribution of tinnitus duration in our study (10% of patients, <1 year; 37.5%, 1 to 2 years; 27.5%, 2 to 5 years; 25%, 5 to 10 years) showed that our study group had duration tinnitus longer than the patients in the study done by the American Tinnitus Association in 1986. This study included 13,000 patients, and the distribution of tinnitus duration was 4.1% of patients < 1 year; 11.7%, 1 to 2 years; 23.1%, 2 to 5 years; 23.3%, 5 to 10 years; 20.7%, 10 to 20 years; and 17.2%, >20 years.¹¹

Some of the reasons for the shorter histories of tinnitus in our study may be advancements in electrophysiology and technology; and better distribution of information on tinnitus, making affected individuals more apt to seek early treatment. Various studies have noted an increased incidence of tinnitus in patients with hearing loss.^{8,12} In tinnitus patients with normal hearing, tinnitus has been associated with diffuse damage to more than 30% of the outer hairy cells, which would not cause any hearing loss.¹³ In our study,

only 4 patients (10%) exhibited medium- or high-level hearing loss in terms of low-frequency air-conduction thresholds with respect to pure tone averages. The theory of Jastreboff and Hazell¹³ may explain why 90% of our patients exhibited no hearing loss in association with tinnitus. These authors consider that tinnitus frequency is generally higher than 2000 Hz, peak at 3000 Hz and 4000 Hz, and is most often recorded close to 4000 Hz.¹ The mean frequency values we measured at initial evaluation were 5796.67 ± 3017.89 Hz (range, 125 to 12000 Hz). The reason for the difference between our tinnitus frequency values and those in the literature may be that we used efficient high-frequency audiometry in our study.

Questionnaires given by the American Tinnitus Association¹¹ have shown that most patients' subjective tinnitus scores are in the 4 to 6 range on a scale of 1 to 10. In our misoprostol-treated group, the mean of the 5 separate measurements varied between 4.9 and 7.2, whereas the corresponding range in the placebo group was 4.6 to 6.6. These findings were in accordance with other results in the literature.

Briner et al.² did a semi-crossover clinical study in an effort to determine how PG analogues affect tinnitus. Misoprostol, which is widely used in gastroenterology and readily available in commercial form, was one of the drugs tested. This is the only published study that has investigated the use of this agent as therapy for tinnitus. The authors started the subjects on 200 μ g/day misoprostol for the first week, and increased by 200 μ g every 5 days to reach a maintenance dose of 800 μ g/day. This latter amount is also the misoprostol dosage commonly used in gastroenterology. In order to calculate the dose-response correlation, the patients underwent daily determination of subjective scores for 1 month. There was no change in the

response rate from days 16 to 20 and days 21 to 25, which was the period in which the medication was given at 600 and 800 $\mu\text{g/day}$. There was a significant difference in the total amount of medication received by the group that responded to treatment and the group that did not respond to treatment. The authors state that cumulative amount of medication may be a significant factor in response to misoprostol therapy. In the group of 12 patients who received misoprostol, 3 patients recovered from tinnitus in less than 1 month. No improvement was observed in 12 other individuals who received placebos, but when misoprostol was subsequently given to this group, 5 of the 12 showed improvement of tinnitus.² Although Briner et al² documented a tinnitus improvement rate of 33%, the reliability of the study has been questioned based on insufficient patient numbers and heterogeneity in the group. Crinnion and McCart¹⁴ listed the following criticisms of the investigation:

1. The characteristics of tinnitus were not indicated.
2. Such small numbers in the sample groups make it difficult to interpret the results.
3. The period effects of misoprostol were not discussed.
4. No specific criteria were used to exclude patients from the study.
5. The study design was not ideal, as the full crossover pattern would have been more appropriate than the semi-crossover pattern.
6. No information was provided regarding the homogeneity of the 2 randomized groups.
7. There was no indication of daily dosage modifications made by patients due to side effects, nor were the final medication dosages reported.
8. There was no mention of diarrhea, the most commonly observed side effect.
9. The effectiveness of misoprostol was evaluated by highly subjective means; a rating scale was used for tinnitus loudness.
10. It was not clarified whether the group that responded to the treatment was standardized (that is, in how many of the medication, placebo, and semi-crossover groups the patient was included).

Crinnion and McCart¹⁴ indicated in their conclusion that the treatment of tinnitus is challenging

because it is difficult to measure tinnitus objectively, and emphasized that larger patient populations must be studied in order to state the effectiveness and reliability of misoprostol as treatment for tinnitus.

In the study by Briner et al,² the researchers asked the question "Do you believe that the medication had an effect in reducing your tinnitus symptoms?" and the answer "Yes" was a subjective criterion for improvement. Based on this, their improvement rate was 33%. In our study, subjective tinnitus scoring was the subjective criterion of improvement that corresponded to this. As explained, we used a drop of 3 grades or more from baseline to the end of 4 months of treatment (that is, the level of annoyance from tinnitus decreased by at least 3 grades on a scale of 1 to 10) as this criterion of improvement. Based on this, our improvement rate was 39% (11 of 28 patients).

Our rates of improvement based on objectively determined tinnitus loudness were 64% at the 15 dB level (18 of 28 patients with reduction of 15 dB or more over 4 months of treatment) and 46% at the 20 dB level (13 of 28 patients with a 20dB or greater improvement in loudness). When we compared these results to findings in the placebo group, we found a significant difference, indicating that misoprostol had a greater positive influence than placebo treatment ($P < 0.05$). However, the considerable level of success achieved with placebos (33% improvement in tinnitus loudness and 17% improvement in subjective tinnitus score) reflects the influence of psychological factors in tinnitus.

Regarding the link between misoprostol-induced improvement and etiological factors, Briner et al² showed that patients with a history of acoustic trauma benefited most from this agent. In contrast, we found that those with a history of sudden onset showed the highest improvement rate, and the acoustic trauma group ranked second.

CONCLUSION

A literature search indicates that our clinical research trial examined misoprostol in the treatment of tinnitus for a period longer than the previous study.² Our study showed that loudness decreased over time (over the 4 months) in the misoprostol group, that each group's subjec-

tive tinnitus scores decreased over this period, that the influence of time on the change of frequency was noticeable, and that the influence of time on the change of subjective tinnitus scores were significant. Our improvement rates were comparable to those in the literature. The misoprostol-treated group showed high improvement rate compared with placebo for 15 dB or more reduction in tinnitus loudness over 4 months. The experimental group also showed moderate improvement rate compared with the placebo group for a reduction in subjective tinnitus score of 3 or more grades over 4 months. The improvement rates for tinnitus loudness and scores of placebo group, reflecting the significance of psychological factors in tinnitus. Our analysis of misoprostol efficacy relative to etiological factors revealed the highest rate of improvement in individuals who had experienced sudden onset of tinnitus. Those with a history of acoustic trauma ranked second in this regard.

In conclusion, our study indicates that misoprostol is a safe treatment that reduces tinnitus loudness (loudness match-dB SL) and the irritation that tinnitus causes. However, further research should focus on larger groups of subjects. It will be important to more deeply investigate the effects of longer term use of misoprostol, as well as the lasting consequences posttreatment.

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